KWF TEAM TOUCANET · FALL 2026

Does the forest react to people?

Detecting human presence from the soundscape's response, not from hearing the people, using the existing AudioMoth recordings.

Bita Massoudi

FOREST RESPONSE · PROGRESS

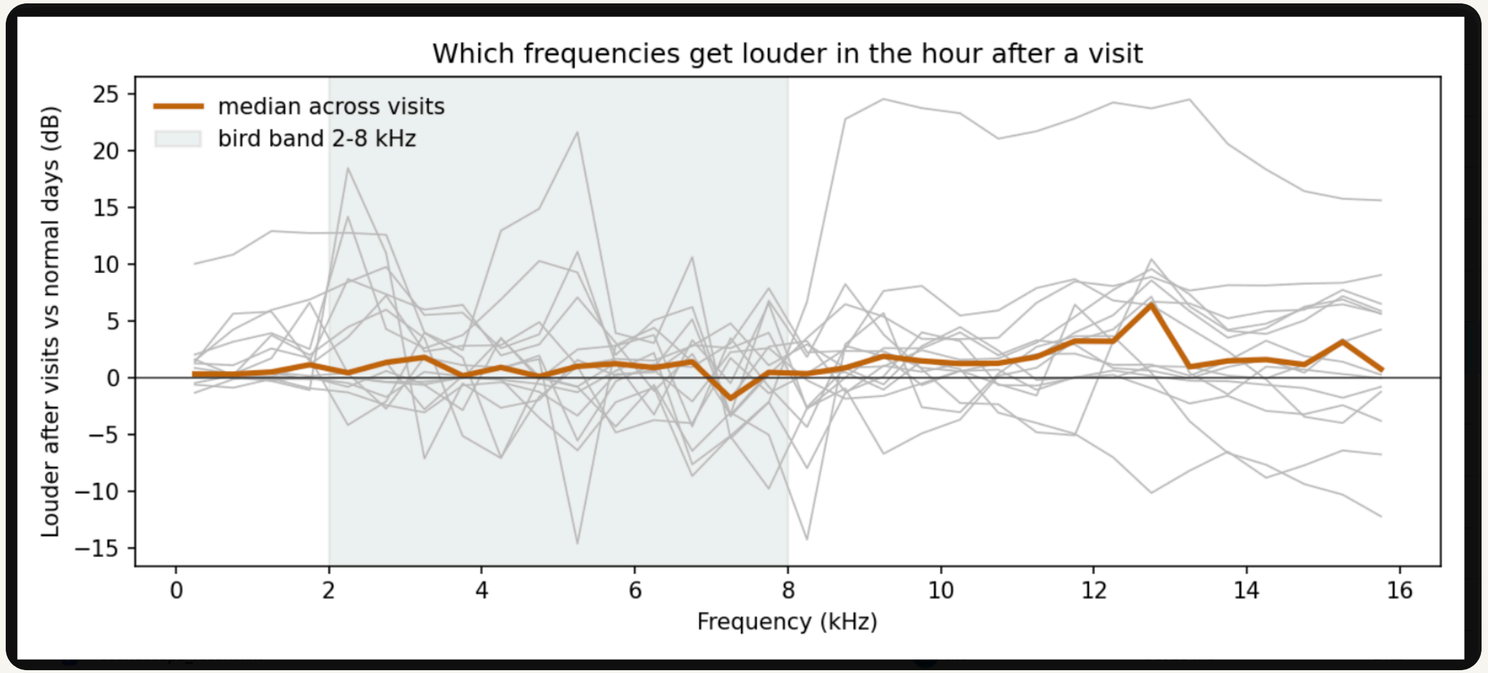
What I tested, step by step

#	Step	Result
1	**Right data**	631k clips, every 3 s; each visit lined up on the moment it was logged in the field
2	**BirdNET, 6 recorders**	Birds heard as often around visits as on normal days; a drop of half or more is ruled out
3	**Raw audio, all 6**	Louder after 23 of 27 visits, at 5 of 6 recorders; not significant after correction
4	**What the extra sound is**	Mostly an insect chorus at 12–13 kHz, and it follows the weather
5	**Exact logged timing**	6–8 kHz a little quieter before visits, in 9 of 15 ($p = 0.07$): a weak hint
6	**Gunshot check**	Our measures pick up the sounds and the people; no clear animal reaction
7	**Model that can't hear people**	Sound above 2 kHz only, no voice or engine clips: recognises unseen visits (AUC 0.69)
8	**Wind check**	Quieter from 40 min before; visit hours ~2 km/h windier
9	**Far hours, same day**	**Birds normal far from visits, though just as windy: the quiet is tied to people**

So far: birds and insects go quieter when people are in the area, even to a model that can't hear people, and not at other hours of the same windy day. Strong evidence, not yet proof.

FOREST RESPONSE · WHAT IS THE EXTRA SOUND?

Mostly insects, not birds



12–13 kHz

Up to +6 dB, louder after 13 of 15 visits (p = 0.007). Narrow and high: an insect chorus.

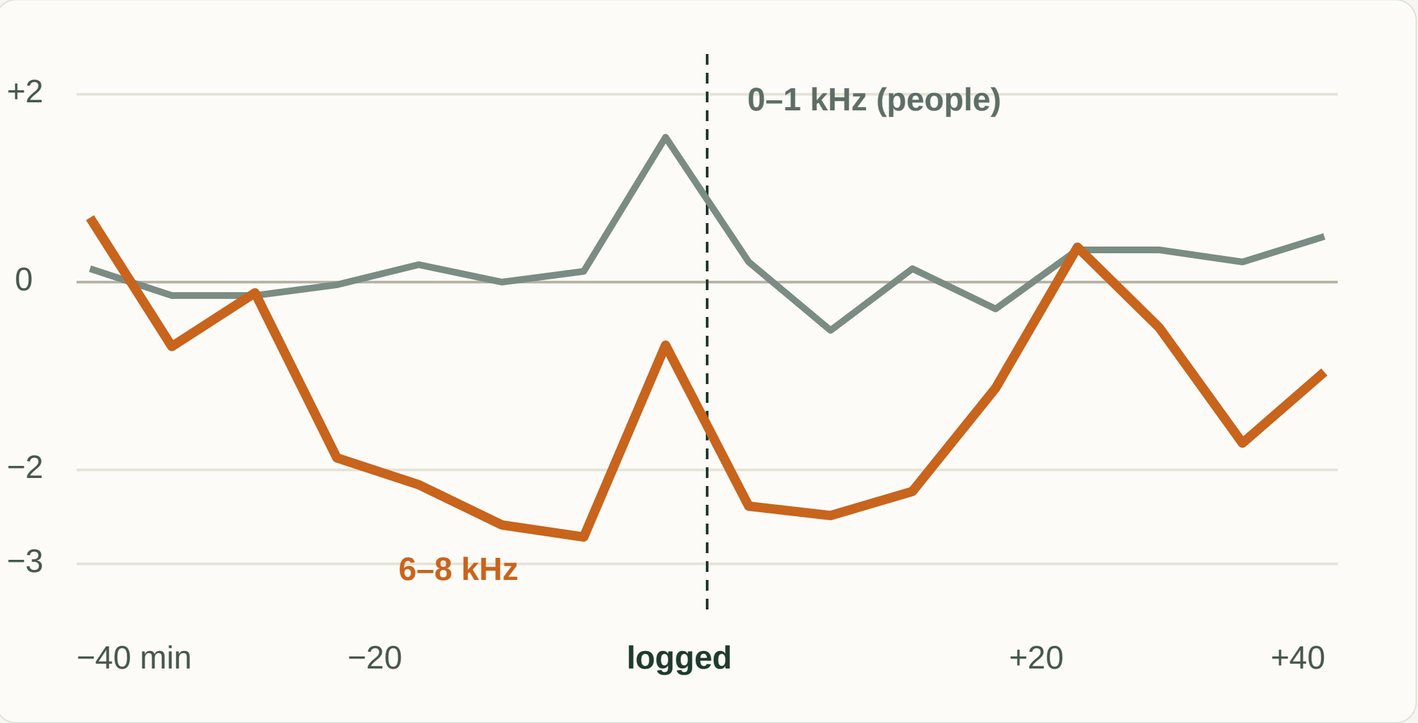
Bird band

Only about +1 dB and patchy; BirdNET finds no extra birds or species.

Weather check: those hours were warmer and drier than normal days, and the rise tracks humidity, so this is probably at least partly weather.

FOREST RESPONSE · AROUND THE LOGGED MOMENT

A weak early-warning hint



6–8 kHz quiets

About 2–2.7 dB from ~25 min before the logged moment; back ~20 min after.

People heard late

The human band rises only in the last 5 min before logging.

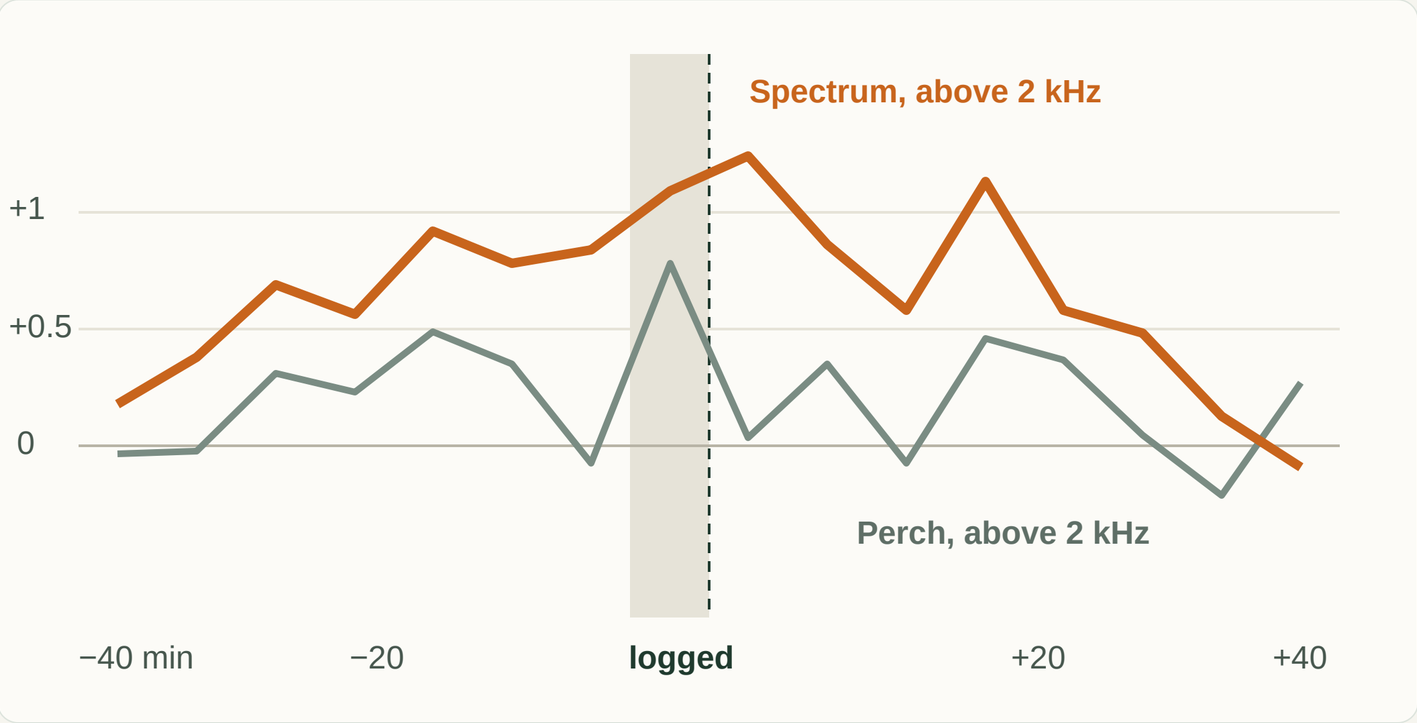
Still weak

Quieter in only 9 of 15 visits (p = 0.07); tinamou not confirmed.

15 human-presence visits, aligned on the logged moment; change vs 40–30 min before, beyond the same recorder's normal days (median dB).

NEW TEST · A MODEL THAT CANNOT HEAR PEOPLE

The forest sounds different around visits



Hears the forest

Recognises visits it never saw (AUC 0.69) with only sound above 2 kHz and no voice or engine clips.

Raised all hour

Up in 10–12 of 15 visits from ~30 min before; it does not wait for the people.

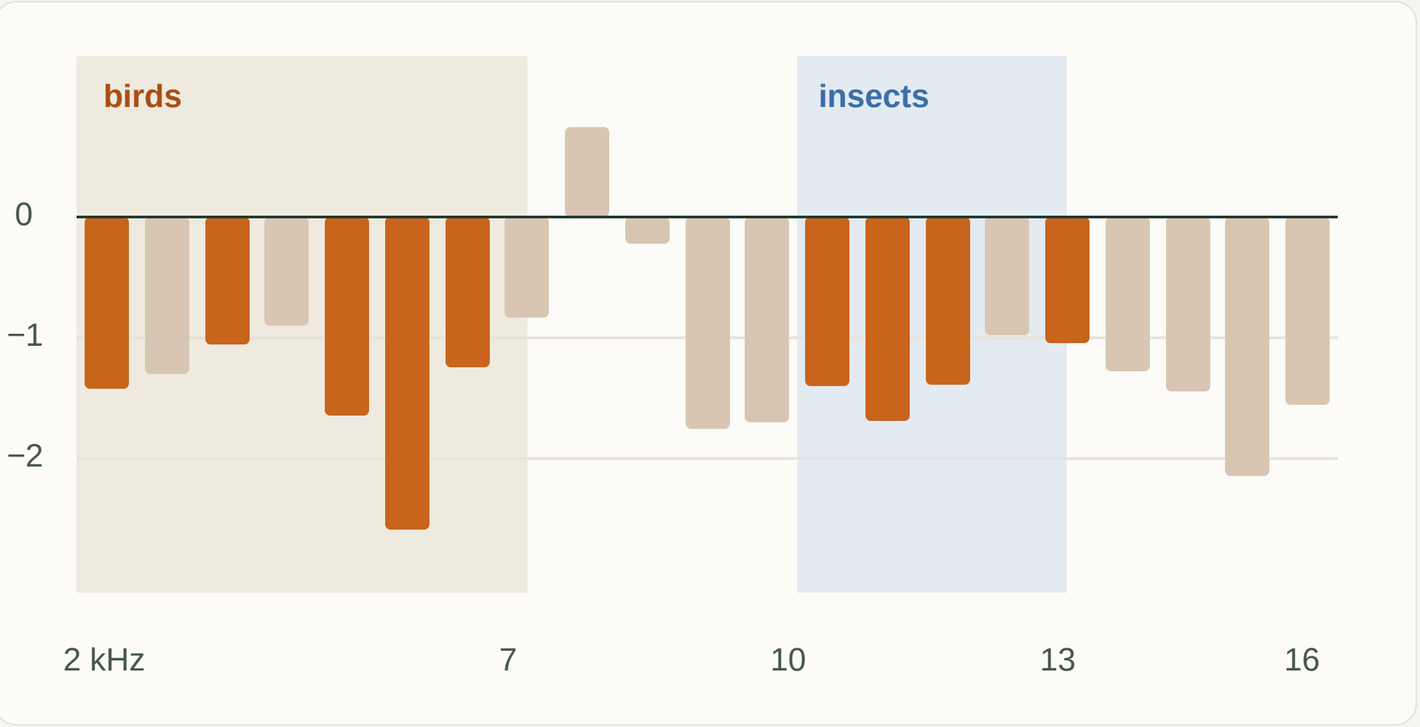
Not yet confirmed

After correcting for 8 tests, best p = 0.08. Perch did worse (AUC 0.59).

Leave-one-visit-out, 15 human-presence visits; score = visit minus its own normal days (same recorder, same clock time), log-odds; grey = people audible.

WHAT THE MODEL HEARS · 2–16 KHZ AROUND VISITS

Birds and insects go quieter



Quieter, not louder

1–2.5 dB quieter in 11–13 of 15 visits, in the bird and insect bands; dark bars agree in ≥ 75%.

Not heat or humidity

Same as normal days at visit hours; still quieter in all 15 after a rough weather adjustment.

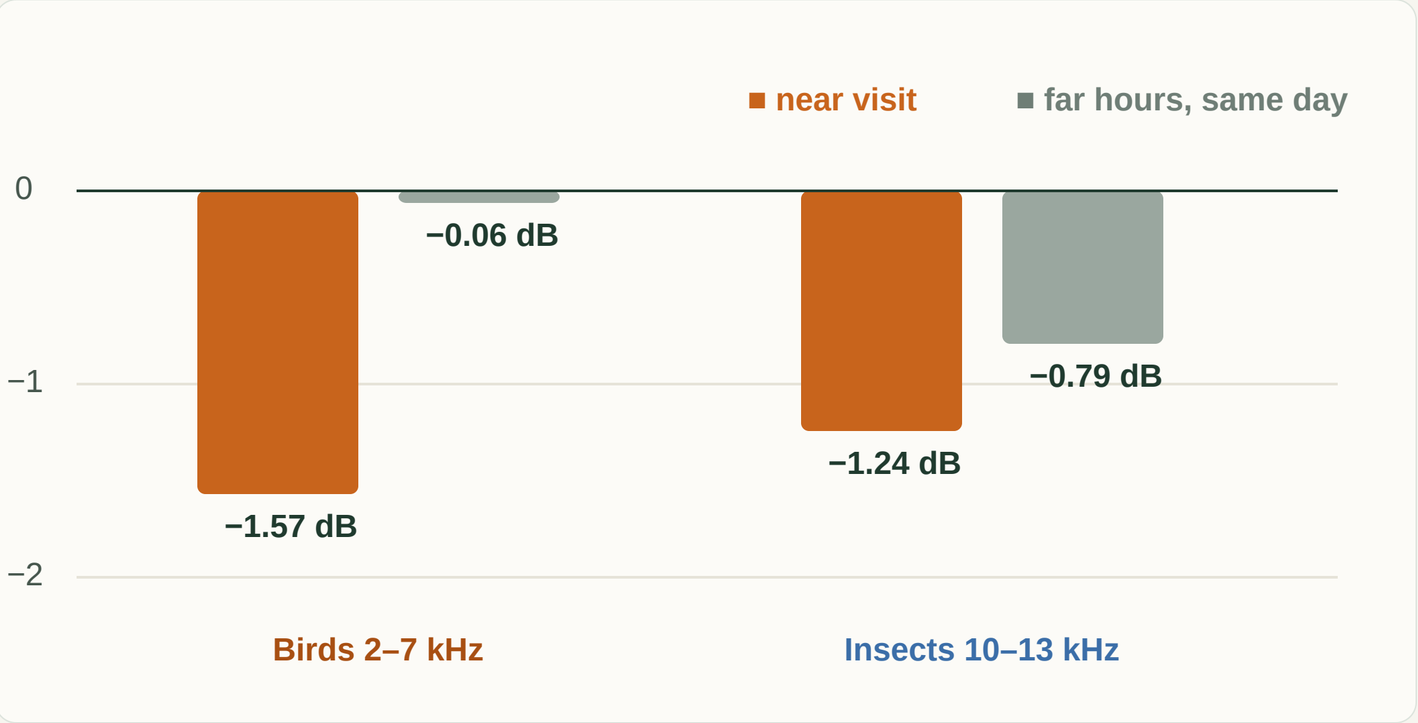
Wind check

Visit hours were ~2 km/h windier, and wind can quiet animals. The next slide tests this directly.

Visit minus normal days, −30 to +30 min around the logged moment, 15 visits, human-tagged clips removed (median dB per 2/3-kHz band).

THE DECIDING TEST · PEOPLE OR THE DAY?

Quiet near visits, not the rest of the day



Quieter near visits only

Same day: quieter near the visit than at far hours. Birds 10 of 13 (p = 0.01), insects 9 of 13 (p = 0.04).

Far hours: birds normal

Hours 3 h or more from any visit are no quieter than normal days (7 of 13).

Not the wind

Far hours were just as windy (+2.3 km/h), yet birds were normal there.

Visit day minus the same clock time on its normal days (median dB); far hours = 3 h or more from any event at that recorder, 2 h from any event anywhere; 13 visit days.

FOREST RESPONSE · SUMMARY

Where we are, and what would settle it

What we found

Birds and insects go quieter when people are in the area, not at other hours of the same day. A model that can't hear people picks it up (AUC 0.69).

Why it's not proof

15 visits, many tests: not significant after correction. The quiet starts ~40 min before logging, so we can't yet tell a reaction from early warning.

What would settle it

Walking-only visits with logged times, on days with similar weather, plus a recorder left undisturbed as a control. Played sounds without people nearby.

Next with current data: get the times the team entered the forest, to see when the quiet starts, and whether any of it comes before people are near.